Adaptive Gaze-Based Focus System

Complete Source Code Documentation

React • TypeScript • MediaPipe FaceMesh • Tailwind CSS

Live Demo: https://gaze-focus-play.lovable.app

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FILE 01

index.html

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <!-- TODO: Set the document title to the name of your application -->
    <title>Lovable App</title>
    <meta name="description" content="An interactive web demo simulating gaze-based focus adjustment using webcam head tracking.">
    <meta name="author" content="Lovable" />

    <!-- TODO: Update og:title to match your application name -->


    <meta property="og:type" content="website" />


    <meta name="twitter:card" content="summary_large_image" />
    <meta name="twitter:site" content="@Lovable" />

    <meta property="og:title" content="Lovable App">
  <meta name="twitter:title" content="Lovable App">
  <meta property="og:description" content="An interactive web demo simulating gaze-based focus adjustment using webcam head tracking.">
  <meta name="twitter:description" content="An interactive web demo simulating gaze-based focus adjustment using webcam head tracking.">
</head>

  <body>
    <div id="root">></div>
    <script type="module" src="/src/main.tsx"></script>
  </body>
</html>

```

FILE 02

package.json

```
{
  "name": "vite_react_shadcn_ts",
  "private": true,
  "version": "0.0.0",
  "type": "module",
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "build:dev": "vite build --mode development",
    "lint": "eslint .",
    "preview": "vite preview",
    "test": "vitest run",
    "test:watch": "vitest"
  },
  "dependencies": {
    "@hookform/resolvers": "^3.10.0",
    "@radix-ui/react-accordion": "^1.2.11",
    "@radix-ui/react-alert-dialog": "^1.1.14",
    "@radix-ui/react-aspect-ratio": "^1.1.7",
    "@radix-ui/react-avatar": "^1.1.10",
    "@radix-ui/react-checkbox": "^1.3.2",
    "@radix-ui/react-collapsible": "^1.1.11",
    "@radix-ui/react-context-menu": "^2.2.15",
    "@radix-ui/react-dialog": "^1.1.14",
    "@radix-ui/react-dropdown-menu": "^2.1.15",
    "@radix-ui/react-hover-card": "^1.1.14",
    "@radix-ui/react-label": "^2.1.7",
    "@radix-ui/react-menubar": "^1.1.15",
    "@radix-ui/react-navigation-menu": "^1.2.13",
    "@radix-ui/react-popover": "^1.1.14",
    "@radix-ui/react-progress": "^1.1.7",
    "@radix-ui/react-radio-group": "^1.3.7",
    "@radix-ui/react-scroll-area": "^1.2.9",
    "@radix-ui/react-select": "^2.2.5",
    "@radix-ui/react-separator": "^1.1.7",
    "@radix-ui/react-slider": "^1.3.5",
    "@radix-ui/react-slot": "^1.2.3",
    "@radix-ui/react-switch": "^1.2.5",
    "@radix-ui/react-tabs": "^1.1.12",
    "@radix-ui/react-toast": "^1.2.14",
    "@radix-ui/react-toggle": "^1.1.9",
    "@radix-ui/react-toggle-group": "^1.1.10",
    "@radix-ui/react-tooltip": "^1.2.7",
    "@tanstack/react-query": "^5.83.0",
    "class-variance-authority": "^0.7.1",
    "clsx": "^2.1.1",
    "cmdk": "^1.1.1",
    "date-fns": "^3.6.0",
    "embla-carousel-react": "^8.6.0",
    "input-otp": "^1.4.2",
    "lucide-react": "^0.462.0",
    "next-themes": "^0.3.0",
    "react": "^18.3.1",
    "react-day-picker": "^8.10.1",
    "react-dom": "^18.3.1",
```

```
    "react-hook-form": "^7.61.1",
    "react-resizable-panels": "^2.1.9",
    "react-router-dom": "^6.30.1",
    "recharts": "^2.15.4",
    "sonner": "^1.7.4",
    "tailwind-merge": "^2.6.0",
    "tailwindcss-animate": "^1.0.7",
    "vaul": "^0.9.9",
    "zod": "^3.25.76"
  },
  "devDependencies": {
    "@eslint/js": "^9.32.0",
    "@tailwindcss/typography": "^0.5.16",
    "@testing-library/jest-dom": "^6.6.0",
    "@testing-library/react": "^16.0.0",
    "@types/node": "^22.16.5",
    "@types/react": "^18.3.23",
    "@types/react-dom": "^18.3.7",
    "@vitejs/plugin-react-swc": "^3.11.0",
    "autoprefixer": "^10.4.21",
    "eslint": "^9.32.0",
    "eslint-plugin-react-hooks": "^5.2.0",
    "eslint-plugin-react-refresh": "^0.4.20",
    "globals": "^15.15.0",
    "jsdom": "^20.0.3",
    "lovable-tagger": "^1.1.13",
    "postcss": "^8.5.6",
    "tailwindcss": "^3.4.17",
    "typescript": "^5.8.3",
    "typescript-eslint": "^8.38.0",
    "vite": "^5.4.19",
    "vitest": "^3.2.4"
  }
}

```

FILE 03

vite.config.ts

```
import { defineConfig } from "vite";
import react from "@vitejs/plugin-react-swc";
import path from "path";
import { componentTagger } from "lovable-tagger";

// https://vitejs.dev/config/
export default defineConfig(({ mode }) => ({
  server: {
    host: "::",
    port: 8080,
    hmr: {
      overlay: false,
    },
  },
  plugins: [react(), mode === "development" && componentTagger()].filter(Boolean),
  resolve: {
    alias: {
      "@": path.resolve(__dirname, "./src"),
    },
    dedupe: ["react", "react-dom", "react/jsx-runtime", "react/jsx-dev-runtime", "@tanstack/react-query", "@tanstack/query-core"],
  },
}));

```

FILE 04

tailwind.config.ts

```
import type { Config } from "tailwindcss";

export default {
  darkMode: ["class"],
  content: ["./pages/**/*.{ts,tsx}", "./components/**/*.{ts,tsx}", "./app/**/*.{ts,tsx}", "./src/**/*.{ts,tsx}"],
  prefix: "",
  theme: {
    container: {
      center: true,
      padding: "2rem",
      screens: {
        "2xl": "1400px",
      },
    },
    extend: {
      colors: {
        border: "hsl(var(--border))",
        input: "hsl(var(--input))",
        ring: "hsl(var(--ring))",
        background: "hsl(var(--background))",
        foreground: "hsl(var(--foreground))",
        primary: {
          DEFAULT: "hsl(var(--primary))",
          foreground: "hsl(var(--primary-foreground))",
        },
        secondary: {
          DEFAULT: "hsl(var(--secondary))",
          foreground: "hsl(var(--secondary-foreground))",
        },
        destructive: {
          DEFAULT: "hsl(var(--destructive))",
          foreground: "hsl(var(--destructive-foreground))",
        },
        muted: {
          DEFAULT: "hsl(var(--muted))",
          foreground: "hsl(var(--muted-foreground))",
        },
        accent: {
          DEFAULT: "hsl(var(--accent))",
          foreground: "hsl(var(--accent-foreground))",
        },
        popover: {
          DEFAULT: "hsl(var(--popover))",
          foreground: "hsl(var(--popover-foreground))",
        },
        card: {
          DEFAULT: "hsl(var(--card))",
          foreground: "hsl(var(--card-foreground))",
        },
        sidebar: {
          DEFAULT: "hsl(var(--sidebar-background))",
          foreground: "hsl(var(--sidebar-foreground))",
          primary: "hsl(var(--sidebar-primary))",
          "primary-foreground": "hsl(var(--sidebar-primary-foreground))",
```

```
55            accent: "hsl(var(--sidebar-accent))",
56            "accent-foreground": "hsl(var(--sidebar-accent-foreground))",
57            border: "hsl(var(--sidebar-border))",
58            ring: "hsl(var(--sidebar-ring))",
59          },
60        },
61        borderRadius: {
62          lg: "var(--radius)",
63          md: "calc(var(--radius) - 2px)",
64          sm: "calc(var(--radius) - 4px)",
65        },
66        keyframes: {
67          "accordion-down": {
68            from: {
69              height: "0",
70            },
71            to: {
72              height: "var(--radix-accordion-content-height)",
73            },
74          },
75          "accordion-up": {
76            from: {
77              height: "var(--radix-accordion-content-height)",
78            },
79            to: {
80              height: "0",
81            },
82          },
83        },
84        animation: {
85          "accordion-down": "accordion-down 0.2s ease-out",
86          "accordion-up": "accordion-up 0.2s ease-out",
87        },
88      },
89    },
90    plugins: [require("tailwindcss-animate")],
91  } satisfies Config;
92
```

FILE 05

tsconfig.json

```
{
  "compilerOptions": {
    "allowJs": true,
    "noImplicitAny": false,
    "noUnusedLocals": false,
    "noUnusedParameters": false,
    "paths": {
      "@/*": [
        "./src/*"
      ]
    },
    "skipLibCheck": true,
    "strictNullChecks": false
  },
  "files": [],
  "references": [
    {
      "path": "./tsconfig.app.json"
    },
    {
      "path": "./tsconfig.node.json"
    }
  ]
}
```

FILE 06

tsconfig.app.json

```
{
  "compilerOptions": {
    "allowImportingTsExtensions": true,
    "isolatedModules": true,
    "jsx": "react-jsx",
    "lib": [
      "ES2020",
      "DOM",
      "DOM.Iterable"
    ],
    "module": "ESNext",
    "moduleDetection": "force",
    "moduleResolution": "bundler",
    "noEmit": true,
    "noFallthroughCasesInSwitch": false,
    "noImplicitAny": false,
    "noUnusedLocals": false,
    "noUnusedParameters": false,
    "paths": {
      "@/*": [
        "./src/*"
      ]
    },
    "skipLibCheck": true,
    "strict": false,
    "target": "ES2020",
    "types": [
      "vitest/globals"
    ],
    "useDefineForClassFields": true
  },
  "include": [
    "src"
  ]
}
```

FILE 07

src/main.tsx

```
import { createRoot } from "react-dom/client";
import App from "./App.tsx";
import "./index.css";

createRoot(document.getElementById("root")!).render(<App />);

```

FILE 08

src/App.tsx

```
import { QueryClient, QueryClientProvider } from "@tanstack/react-query";
import { BrowserRouter, Route, Routes } from "react-router-dom";
import { Toaster as Sonner } from "@/components/ui/sonner";
import { Toaster } from "@/components/ui/toaster";
import { TooltipProvider } from "@/components/ui/tooltip";
import Index from "./pages/Index.tsx";
import NotFound from "./pages/NotFound.tsx";

const queryClient = new QueryClient();

const App = () => (
  <QueryClientProvider client={queryClient}>
    <TooltipProvider>
      <Toaster />
      <Sonner />
      <BrowserRouter>
        <Routes>
          <Route path="/" element={<Index />} />
          {/* ADD ALL CUSTOM ROUTES ABOVE THE CATCH-ALL "*" ROUTE */}
          <Route path="*" element={<NotFound />} />
        </Routes>
      </BrowserRouter>
    </TooltipProvider>
  </QueryClientProvider>
);

export default App;

```

FILE 09

src/index.css

```
@tailwind base;
@tailwind components;
@tailwind utilities;

@import
url('https://fonts.googleapis.com/css2?family=JetBrains+Mono:wght@400;500;600;700&family=Space+Grotes
k:wght@300;400;500;600;700&display=swap');

@layer base {
  :root {
    --background: 220 20% 6%;
    --foreground: 200 20% 92%;

    --card: 220 18% 10%;
    --card-foreground: 200 20% 92%;

    --popover: 220 18% 10%;
    --popover-foreground: 200 20% 92%;

    --primary: 174 72% 50%;
    --primary-foreground: 220 20% 6%;

    --secondary: 220 16% 16%;
    --secondary-foreground: 200 20% 80%;

    --muted: 220 14% 14%;
    --muted-foreground: 200 10% 50%;

    --accent: 290 60% 60%;
    --accent-foreground: 200 20% 95%;

    --destructive: 0 72% 55%;
    --destructive-foreground: 210 40% 98%;

    --border: 220 14% 18%;
    --input: 220 14% 18%;
    --ring: 174 72% 50%;

    --radius: 0.75rem;

    --glow-primary: 0 0 20px hsl(174 72% 50% / 0.3);
    --glow-accent: 0 0 20px hsl(290 60% 60% / 0.3);

    --font-display: 'Space Grotesk', sans-serif;
    --font-mono: 'JetBrains Mono', monospace;

    --sidebar-background: 220 18% 8%;
    --sidebar-foreground: 200 20% 92%;
    --sidebar-primary: 174 72% 50%;
    --sidebar-primary-foreground: 220 20% 6%;
    --sidebar-accent: 220 16% 16%;
    --sidebar-accent-foreground: 200 20% 80%;
    --sidebar-border: 220 14% 18%;
    --sidebar-ring: 174 72% 50%;
  }
```

```
}

@layer base {
  * {
    @apply border-border;
  }
  body {
    @apply bg-background text-foreground;
    font-family: var(--font-display);
  }
  code, .font-mono {
    font-family: var(--font-mono);
  }
}

@layer utilities {
  .glow-primary {
    box-shadow: var(--glow-primary);
  }
  .glow-accent {
    box-shadow: var(--glow-accent);
  }
  .glass {
    @apply bg-card/60 backdrop-blur-xl border border-border/50;
  }
  .zone-blur {
    filter: blur(12px) brightness(0.6);
    transition: filter 0.6s cubic-bezier(0.4, 0, 0.2, 1), transform 0.6s cubic-bezier(0.4, 0, 0.2, 1);
  }
  .zone-clear {
    filter: blur(0px) brightness(1);
    transform: scale(1.02);
    transition: filter 0.6s cubic-bezier(0.4, 0, 0.2, 1), transform 0.6s cubic-bezier(0.4, 0, 0.2, 1);
  }
}

```

FILE 10

src/pages/Index.tsx

```
import GazeDemo from "@/components/GazeDemo";

export default function Index() {
  return <GazeDemo />;
}

```

FILE 11

src/pages/NotFound.tsx

```
import { useLocation } from "react-router-dom";
import { useEffect } from "react";

const NotFound = () => {
  const location = useLocation();

  useEffect(() => {
    console.error("404 Error: User attempted to access non-existent route:", location.pathname);
  }, [location.pathname]);

  return (
    <div className="flex min-h-screen items-center justify-center bg-muted">
      <div className="text-center">
        <h1 className="mb-4 text-4xl font-bold">404</h1>
        <p className="mb-4 text-xl text-muted-foreground">Oops! Page not found</p>
        <a href="/" className="text-primary underline hover:text-primary/90">
          Return to Home
        </a>
      </div>
    </div>
  );
};

export default NotFound;

```

FILE 12

src/components/GazeDemo.tsx

```
import { useRef, useEffect, useState } from "react";
import { useGazeTracking, type GazeDirection } from "@/hooks/useGazeTracking";

const ZONE_LABELS: Record<GazeDirection, string> = {
  left: "Looking Left",
  center: "Looking Center",
  right: "Looking Right",
  unknown: "No Face Detected",
};

export default function GazeDemo() {
  const videoRef = useRef<HTMLVideoElement | null>(null);
  const [cameraReady, setCameraReady] = useState(false);
  const [cameraError, setCameraError] = useState("");
  const gaze = useGazeTracking(videoRef);

  useEffect(() => {
    async function startCamera() {
      try {
        const stream = await navigator.mediaDevices.getUserMedia({
          video: { width: 640, height: 480, facingMode: "user" },
        });
        if (videoRef.current) {
          videoRef.current.srcObject = stream;
          await videoRef.current.play();
          setCameraReady(true);
        }
      } catch {
        setCameraError("Camera access denied. Please allow camera access to use this demo.");
      }
    }
    startCamera();
    return () => {
      if (videoRef.current?.srcObject) {
        (videoRef.current.srcObject as MediaStream).getTracks().forEach((t) => t.stop());
      }
    };
  }, []);

  const dir = gaze.faceDetected ? gaze.direction : "unknown";

  const focusX = dir === "left" ? "16.5%" : dir === "right" ? "83.5%" : "50%";

  return (
    <div className="relative w-full h-screen overflow-hidden bg-background">
      {/* Background scene with 3 zones */}
      <div className="absolute inset-0 flex">
        {(["left", "center", "right"] as const).map((zone) => (
          <div
            key={zone}
            className={`flex-1 bg-cover bg-center ${
              dir === zone || dir === "unknown" ? "zone-clear" : "zone-blur"
            }`}
            style={{
              backgroundImage: "url('/images/scene.jpg')",
```

```
              backgroundPosition:
                zone === "left" ? "left center" : zone === "right" ? "right center" : "center",
            }}
          />
        ))}
      </div>

      {/* Radial spotlight overlay */}
      <div
        className="absolute inset-0 pointer-events-none transition-all duration-700 ease-out"
        style={{
          background:
            dir === "unknown"
              ? "transparent"
              : `radial-gradient(ellipse 40% 80% at ${focusX} 50%, transparent 0%, hsl(var(--background) / 0.5) 100%)`,
        }}
      />

      {/* Top HUD */}
      <div className="absolute top-6 left-1/2 -translate-x-1/2 z-20 flex flex-col items-center gap-2">
        <h1 className="text-xl md:text-2xl font-bold tracking-tight text-foreground drop-shadow-lg">
          Adaptive Gaze-Based Focus System
        </h1>
        <p className="text-sm text-muted-foreground font-mono">
          Move your head left / right to change focus
        </p>
      </div>

      {/* Direction label */}
      <div className="absolute top-1/2 left-1/2 -translate-x-1/2 -translate-y-1/2 z-20 pointer-events-none">
        <div className="glass rounded-2xl px-8 py-4 glow-primary text-center">
          <p className="text-2xl md:text-4xl font-bold text-primary tracking-wide">
            {ZONE_LABELS[dir]}
          </p>
        </div>
      </div>

      {/* Bottom panel */}
      <div className="absolute bottom-6 left-1/2 -translate-x-1/2 z-20 flex flex-col items-center gap-3 w-full max-w-xl px-4">
        {/* Stats */}
        <div className="glass rounded-xl px-5 py-3 flex items-center gap-6 font-mono text-xs text-muted-foreground">
          <span>
            FPS: <span className="text-primary font-semibold">{gaze.fps}</span>
          </span>
          <span>
            Confidence:{" "}
            <span className="text-primary font-semibold">
              {(gaze.confidence * 100).toFixed(0)}%
            </span>
          </span>
          <span>
            Face:{" "}
            <span className={gaze.faceDetected ? "text-primary" : "text-destructive"}>
```

```
                {gaze.faceDetected ? "Detected" : "Not Found"}
              </span>
            </span>
            <span>
              Offset:{" "}
              <span className="text-accent font-semibold">{gaze.rawNoseX.toFixed(3)}</span>
            </span>
          </div>

          {/* Instructions & Explanation */}
          <div className="glass rounded-xl px-5 py-3 text-center text-xs text-muted-foreground leading-relaxed max-w-lg">
            This demo simulates solving the{" "}
            <span className="text-accent font-semibold">Vergence-Accommodation Conflict</span> by
            dynamically adjusting focus based on user gaze direction — similar to adaptive varifocal
            rendering in VR/AR headsets.
          </div>
        </div>

        {/* Webcam preview */}
        <div className="absolute bottom-6 right-6 z-20">
          <div className="glass rounded-xl overflow-hidden glow-primary w-40 md:w-52">
            <video
              ref={videoRef}
              className="w-full h-auto mirror"
              style={{ transform: "scaleX(-1)" }}
              playsInline
              muted
            />
            {!cameraReady && !cameraError && (
              <div className="absolute inset-0 flex items-center justify-center text-xs text-muted-foreground">
                Loading camera…
              </div>
            )}
          </div>
        </div>

        {/* Camera error overlay */}
        {cameraError && (
          <div className="absolute inset-0 z-30 flex items-center justify-center bg-background/80">
            <div className="glass rounded-2xl px-8 py-6 max-w-md text-center">
              <p className="text-destructive font-semibold mb-2">Camera Error</p>
              <p className="text-sm text-muted-foreground">{cameraError}</p>
            </div>
          </div>
        )}
      </div>
    );
  }

```

FILE 13

src/hooks/useGazeTracking.ts

```
import { useEffect, useRef, useState, useCallback } from "react";

export type GazeDirection = "left" | "center" | "right" | "unknown";

interface GazeState {
  direction: GazeDirection;
  confidence: number;
  fps: number;
  faceDetected: boolean;
  rawNoseX: number;
}

const SMOOTHING = 0.15;
const LEFT_THRESHOLD = -0.06;
const RIGHT_THRESHOLD = 0.06;
const HYSTERESIS = 0.02;

function loadScript(src: string): Promise<void> {
  return new Promise((resolve, reject) => {
    if (document.querySelector(`script[src="${src}"]`)) {
      resolve();
      return;
    }
    const s = document.createElement("script");
    s.src = src;
    s.crossOrigin = "anonymous";
    s.onload = () => resolve();
    s.onerror = reject;
    document.head.appendChild(s);
  });
}

export function useGazeTracking(videoRef: React.RefObject<HTMLVideoElement | null>) {
  const [state, setState] = useState<GazeState>({
    direction: "center",
    confidence: 0,
    fps: 0,
    faceDetected: false,
    rawNoseX: 0,
  });

  const smoothedX = useRef(0);
  const lastDirection = useRef<GazeDirection>("center");
  const frameCount = useRef(0);
  const lastFpsTime = useRef(Date.now());
  const currentFps = useRef(0);
  const animFrameId = useRef<number>(0);
  const faceMeshRef = useRef<any>(null);

  const detect = useCallback(async () => {
    const video = videoRef.current;
    if (!video || !faceMeshRef.current || video.readyState < 2) {
      animFrameId.current = requestAnimationFrame(detect);
      return;
    }
```

```

    try {
      await faceMeshRef.current.send({ image: video });
    } catch {
      // ignore transient errors
    }

    frameCount.current++;
    const now = Date.now();
    if (now - lastFpsTime.current >= 1000) {
      currentFps.current = frameCount.current;
      frameCount.current = 0;
      lastFpsTime.current = now;
    }

    animFrameId.current = requestAnimationFrame(detect);
  }, [videoRef]);

  useEffect(() => {
    let cancelled = false;

    async function init() {
      await loadScript(
        "https://cdn.jsdelivr.net/npm/@mediapipe/face_mesh@0.4.1633559619/face_mesh.js"
      );

      if (cancelled) return;

      const win = window as any;
      const FaceMesh = win.FaceMesh;
      if (!FaceMesh) return;

      const faceMesh = new FaceMesh({
        locateFile: (file: string) =>
          `https://cdn.jsdelivr.net/npm/@mediapipe/face_mesh@0.4.1633559619/${file}`,
      });

      faceMesh.setOptions({
        maxNumFaces: 1,
        refineLandmarks: false,
        minDetectionConfidence: 0.5,
        minTrackingConfidence: 0.5,
      });

      faceMesh.onResults((results: any) => {
        if (!results.multiFaceLandmarks || results.multiFaceLandmarks.length === 0) {
          setState((s) => ({ ...s, faceDetected: false, confidence: 0, fps: currentFps.current
}));
          return;
        }

        const landmarks = results.multiFaceLandmarks[0];
        const noseTip = landmarks[1];
        const leftCheek = landmarks[234];
        const rightCheek = landmarks[454];
        const faceCenter = (leftCheek.x + rightCheek.x) / 2;
        const faceWidth = Math.abs(rightCheek.x - leftCheek.x);

        const offset = (noseTip.x - faceCenter) / (faceWidth || 1);
```

```

        smoothedX.current += (offset - smoothedX.current) * SMOOTHING;
        const sx = smoothedX.current;

        let dir: GazeDirection = lastDirection.current;
        if (dir === "center") {
          if (sx < LEFT_THRESHOLD) dir = "left";
          else if (sx > RIGHT_THRESHOLD) dir = "right";
        } else if (dir === "left") {
          if (sx > LEFT_THRESHOLD + HYSTERESIS) dir = "center";
        } else if (dir === "right") {
          if (sx < RIGHT_THRESHOLD - HYSTERESIS) dir = "center";
        }

        lastDirection.current = dir;

        const conf = Math.min(1, Math.abs(sx) / 0.15);

        setState({
          direction: dir,
          confidence: dir === "center" ? 1 - conf : conf,
          fps: currentFps.current,
          faceDetected: true,
          rawNoseX: sx,
        });
      });

      faceMeshRef.current = faceMesh;
      animFrameId.current = requestAnimationFrame(detect);
    }

    init();

    return () => {
      cancelled = true;
      cancelAnimationFrame(animFrameId.current);
      if (faceMeshRef.current) {
        faceMeshRef.current.close();
      }
    };
  }, [detect]);

  return state;
}

```

FILE 14

src/lib/utils.ts

```
import { clsx, type ClassValue } from "clsx";
import { twMerge } from "tailwind-merge";

export function cn(...inputs: ClassValue[]) {
  return twMerge(clsx(inputs));
}

```